

Survival Model Evaluation: `chicago_licences`

Fitted with the strategy-survival-model pipeline; methodology validated separately against synthetic ground truth

Source data	<code>datasets/chicago_licences.csv.gz</code>
Rows	342,453 (84.6% with the ending observed, 15.4% censored)
Start dates	2002-01-02 to 2026-07-31
Evaluation	5 expanding-window temporal folds
Source of figures	<code>reports/chicago_demo/metrics.json</code> , regenerated by the command in section 6

1. Summary

This report evaluates a survival model fitted to `chicago_licences.csv.gz` : 342,453 rows, each observed from its start date for 'licensed_days' days with 'closed' marking whether the ending was seen (84.6%) or the row was still running when observation stopped (15.4%). Median observed duration is 977 days. The model predicts, from the 173 features available at the start date, how long each row survives.

Pooled over 205,472 out-of-time test rows, the XGBoost AFT model reaches a concordance index of 0.665 (95% bootstrap interval 0.663 to 0.666), against 0.671 for a Cox proportional hazards baseline on the same features and 0.500 for a coin flip. At the 1095-day horizon its censoring-weighted Brier score is 0.216, beating the 0.249 of a no-skill forecast that assigns every row the population average.

This dataset has no known generating process. Unlike the synthetic validation report, there is no oracle ceiling to say how much signal remains unclaimed, and feature attributions cannot be checked against a true mechanism. What carries over from the synthetic validation is the pipeline itself: the same temporal folds, label re-censoring, likelihood-based selection, and calibration checks, verified there against known answers.

2. Method

Rows are ordered by start date and evaluated with 5 expanding-window folds; every fold trains only on rows that started before its split date and tests on the next block, so training sets grow from 136,978 to 301,356 rows. Training labels are re-censored at each split date: an ending recorded after the split is rewritten to "still running at the split", which is all a model trained then could have known. Test labels keep their full outcomes.

The model is XGBoost with the `survival:aft` objective; censoring enters through interval labels. Hyperparameters are selected on the first fold's training window by held-out censored log-likelihood (selected depth 3, loss scale 1.2), and the predictive log-normal scale is calibrated separately on a temporal tail slice (final value 1.28). Numeric features pass through as-is, missing values included, which XGBoost handles natively; the Cox baseline receives train-window median imputation. Text columns are one-hot encoded.

3. Results

Table 1. Concordance index, pooled across folds. Higher is better; 0.500 is a coin flip.

METHOD	HARRELL C	95% INTERVAL
XGBoost AFT (pooled)	0.665	0.663 to 0.666
XGBoost AFT (fold mean)	0.663	not resampled
Cox proportional hazards (fold mean)	0.671	not resampled

On this dataset the Cox baseline outscores the boosted model, 0.671 against 0.663 on fold-mean concordance. Both fitted models are saved and the Cox model is the one this run recommends for scoring new rows. Reporting that, rather than presenting the boosted model as the result, is the point of carrying a baseline at all. Cox never estimates a baseline hazard, so its scores are only meaningful within a fold; the fold-mean rows are the like-for-like comparison and the pooled figure is not comparable to them.

Table 2. Per-fold results. Censoring is the share of each fold's test rows whose ending was not observed.

FOLD	SPLIT DATE	TRAIN N	TEST N	CENSORED	AFT	COX
1	2007-02-21	136,978	41,095	5%	0.556	0.584
2	2010-04-02	178,047	41,095	8%	0.675	0.672
3	2012-12-29	211,797	41,094	15%	0.640	0.650
4	2016-06-24	260,257	41,094	24%	0.684	0.687
5	2021-03-30	301,356	41,094	57%	0.760	0.759

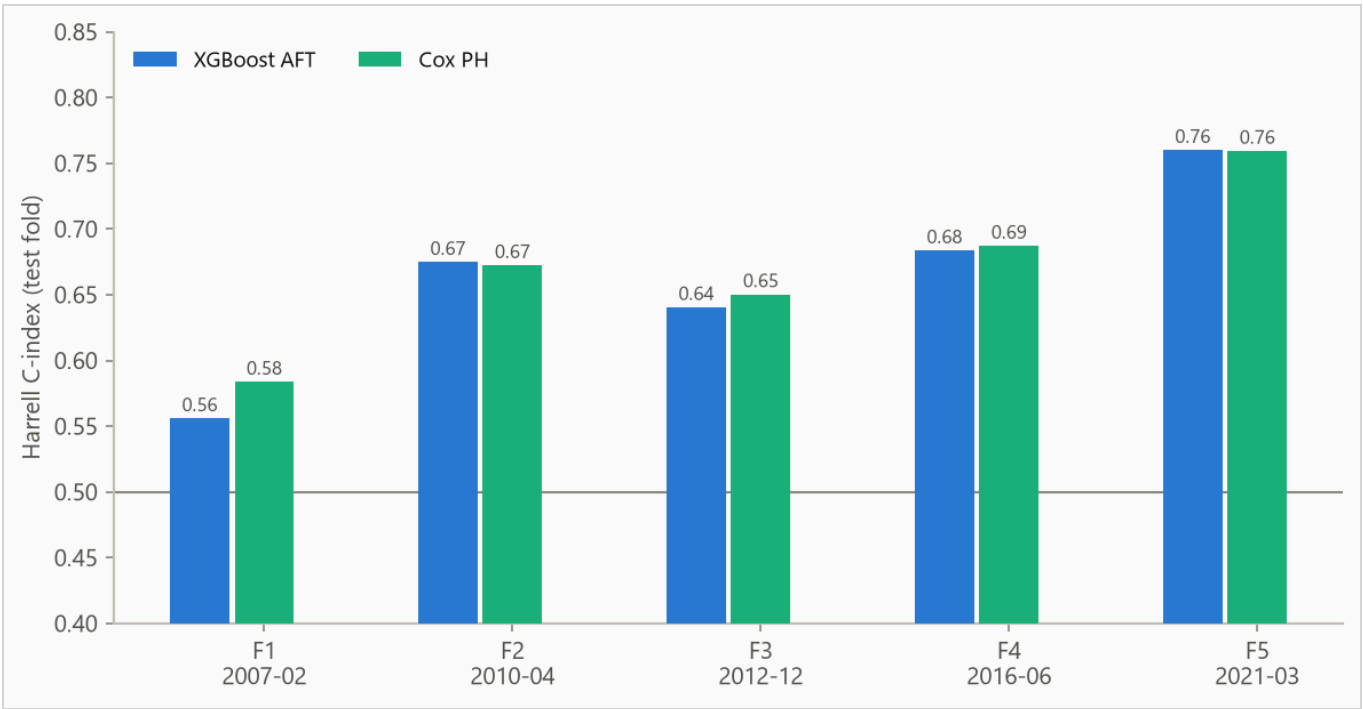


Figure 1. Concordance by fold, from 0.556 to 0.760. Folds differ in censoring mix, so cross-fold comparisons are indicative rather than exact.

Table 3. IPCW Brier score by horizon; lower is better. The no-skill column assigns every row the pooled Kaplan-Meier marginal.

HORIZON	AFT	COX	NO-SKILL MARGINAL
365 days	0.100	0.099	0.153
1095 days	0.216	0.212	0.249
1825 days	0.196	0.193	0.215

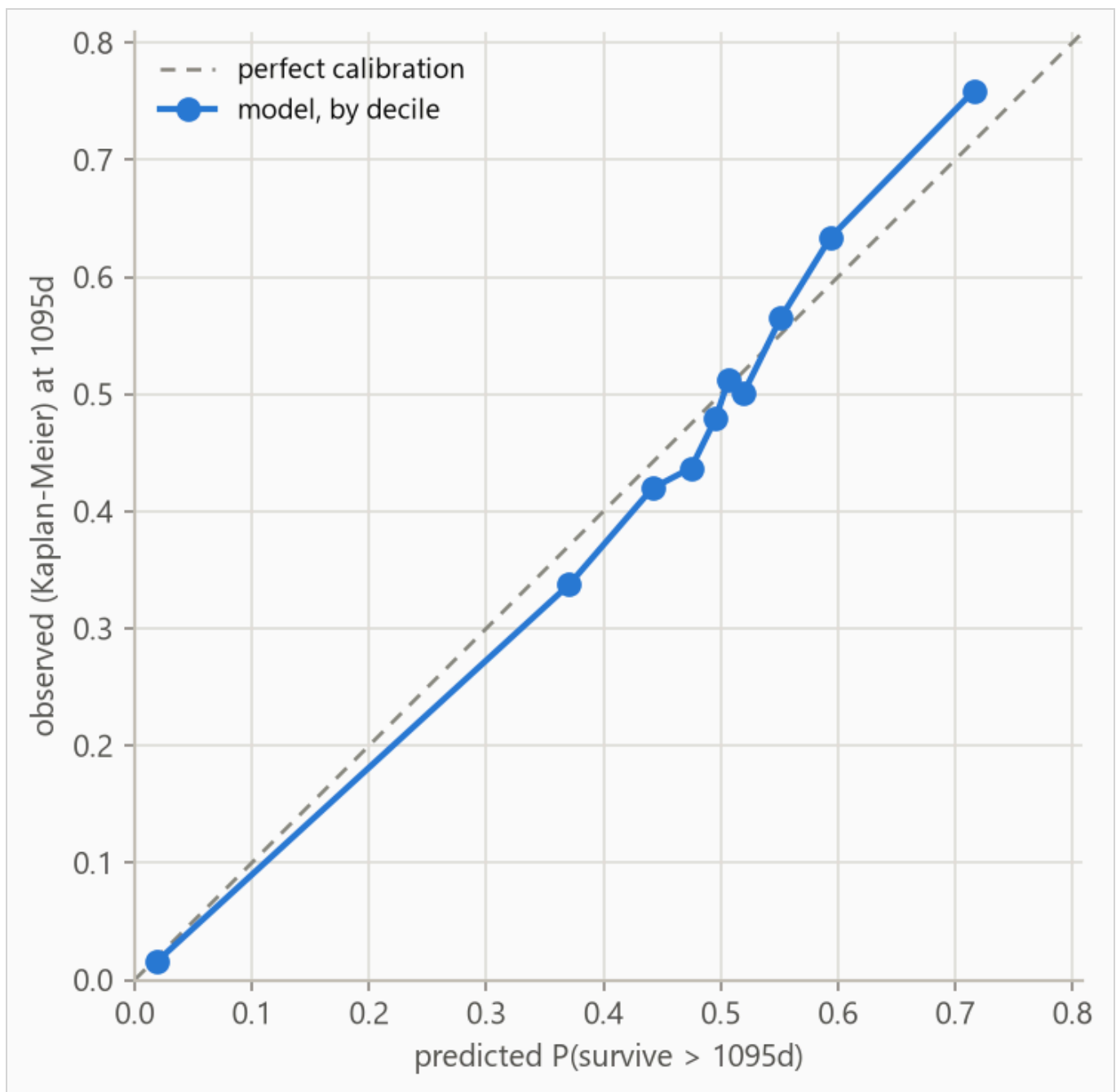


Figure 2. Predicted against observed survival at 1095 days, by predicted decile. Observed frequencies are Kaplan-Meier estimates within each bin, so censored rows contribute correctly.

Table 4. Decile calibration at 1095 days, the values plotted in Figure 2.

DECILE	N	PREDICTED	OBSERVED (KM)	DEVIATION
1	20549	0.019	0.015	-0.003
2	20546	0.371	0.338	-0.033
3	20566	0.442	0.420	-0.022
4	21841	0.475	0.437	-0.038
5	19272	0.496	0.480	-0.016
6	21025	0.507	0.513	+0.006
7	20032	0.519	0.501	-0.019
8	20952	0.551	0.565	+0.014
9	20155	0.594	0.634	+0.040
10	20534	0.716	0.759	+0.043

4. What the model uses

Feature attributions are computed on the log-time margin: positive values push predicted survival up. On this data they are descriptive only. There is no known mechanism to check them against, and correlated features can split credit in ways that reflect the model's internal choices as much as the data, so read the ranking as "what this model leaned on", not as importance in the world.

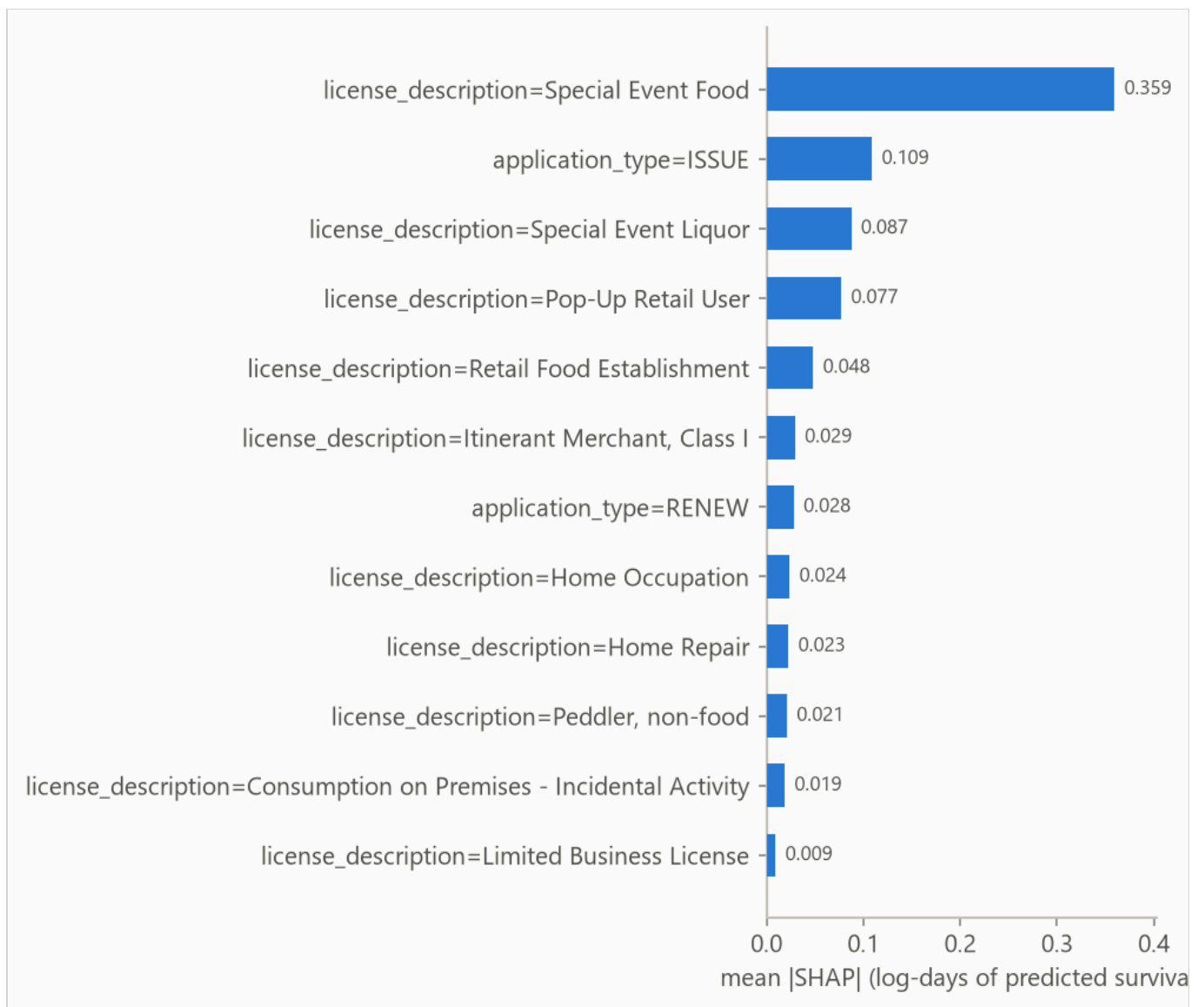


Figure 3. Mean absolute attribution by feature.

Table 5. Top eight features by mean absolute attribution.

FEATURE	MEAN ATTRIBUTION
license_description=Special Event Food	0.359
application_type=ISSUE	0.109
license_description=Special Event Liquor	0.087
license_description=Pop-Up Retail User	0.077
license_description=Retail Food Establishment	0.048
license_description=Itinerant Merchant, Class I	0.029
application_type=RENEW	0.028
license_description=Home Occupation	0.024

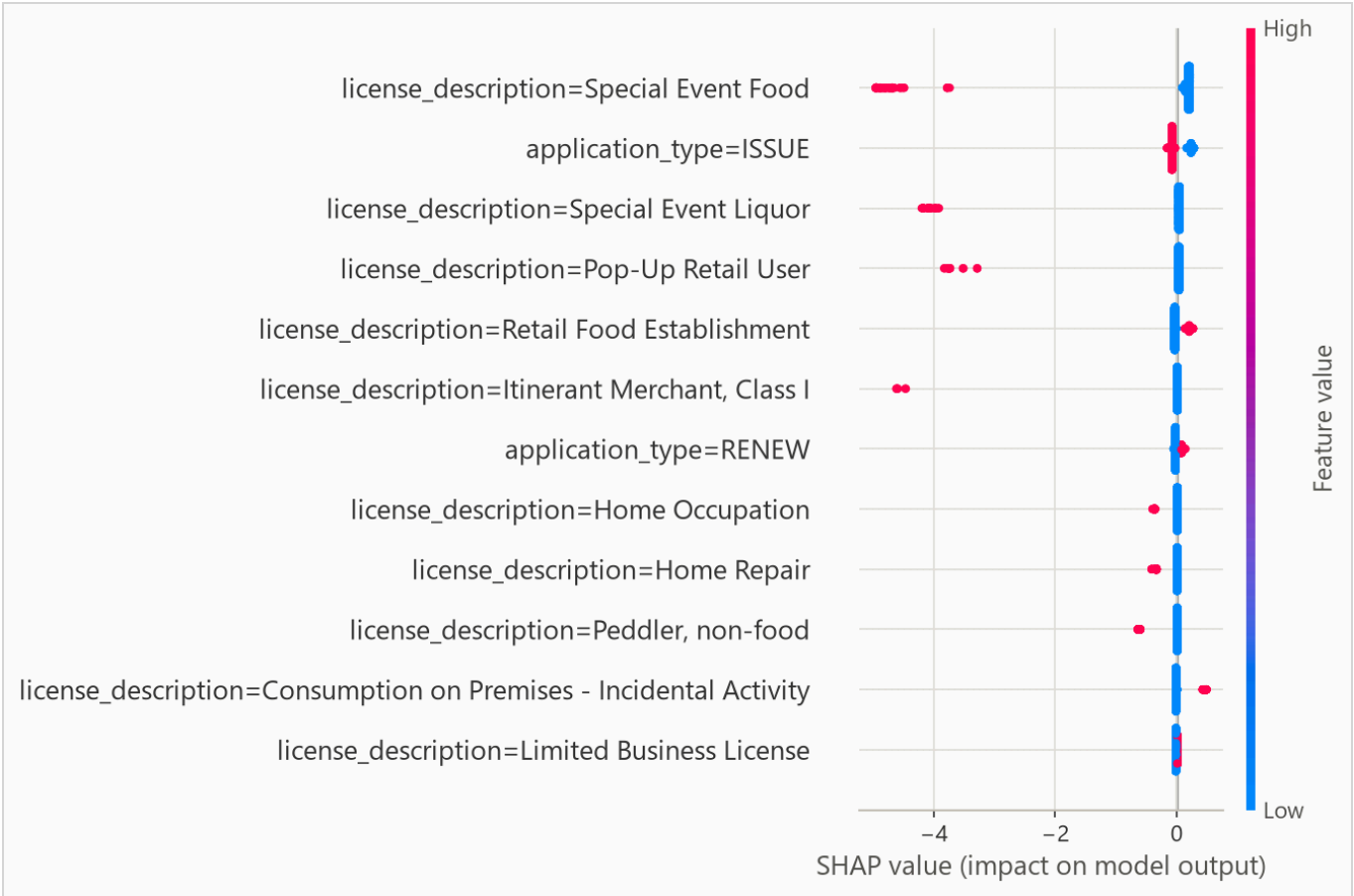


Figure 4. Per-row attributions across the explanation sample.

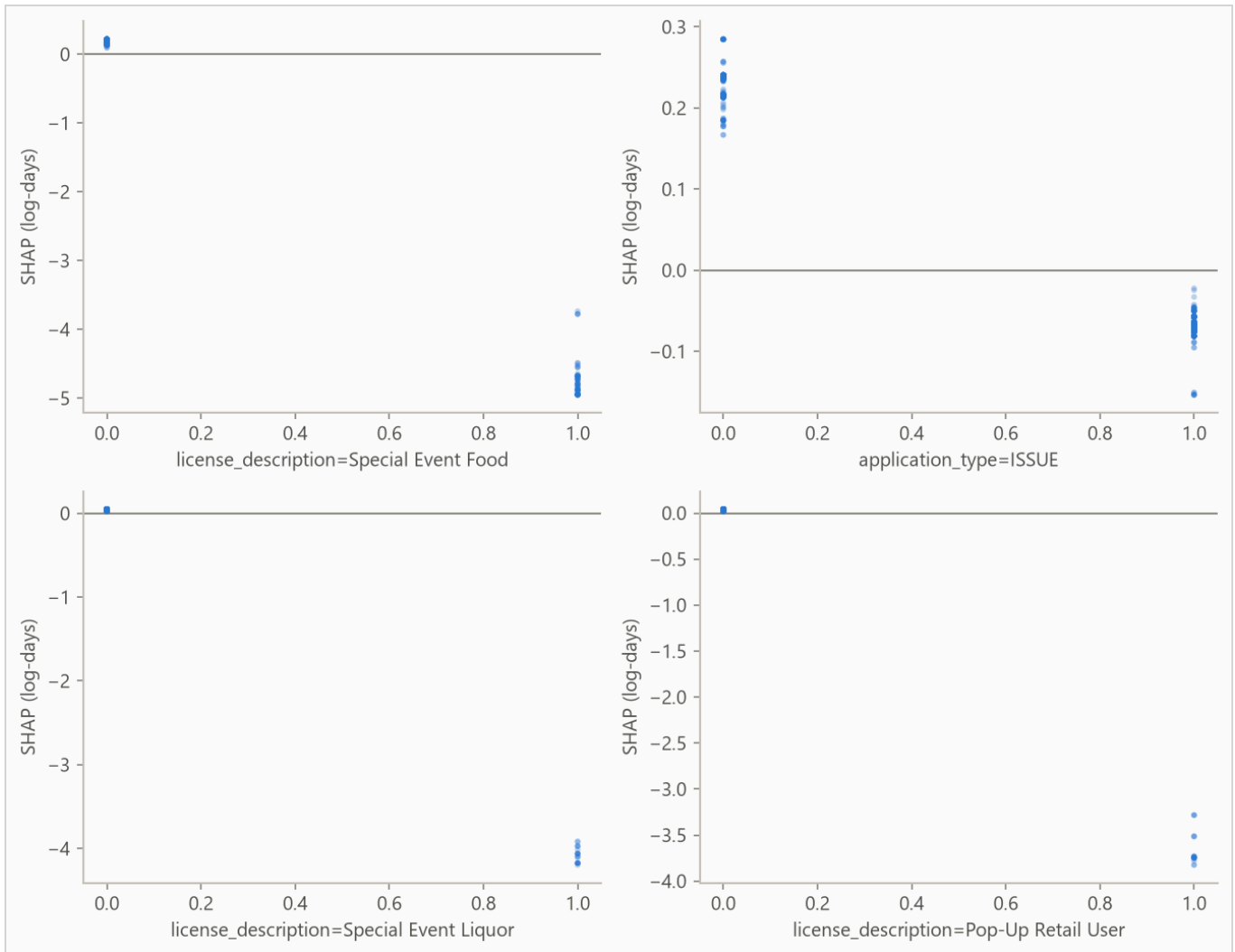


Figure 5. Attribution against feature value for the four features with the largest mean attribution.

5. Limitations

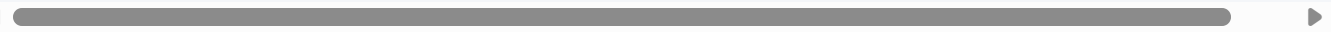
No ground truth. Nothing bounds how much predictable signal this dataset contains, so the concordance above cannot be placed relative to a ceiling the way the synthetic validation could.

Censoring may be informative. The evaluation assumes rows stop being observed for reasons unrelated to their risk. If observation ended early on rows that were about to end anyway, survival estimates are biased upward. Whether that holds here depends on how this dataset was collected.

Harrell's concordance is biased under heavy censoring. Fold-to-fold comparisons in Table 2 should be read with each fold's censoring share in mind.

6. Reproducing this run

```
python scripts/run_fit_evaluate.py --data datasets/chicago_licences.csv.gz --name chicago_licences
    --id-col licence_id --date-col first_issued ^
    --duration-col licensed_days --event-col closed --folds 5 --horizons 365,1095,1825
python scripts/run_build_report.py --run reports/chicago_demo
```



Generated from `reports/chicago_demo/metrics.json` by `scripts/run_build_report.py --run`. 342,453 rows,
205,472 out-of-time test rows.